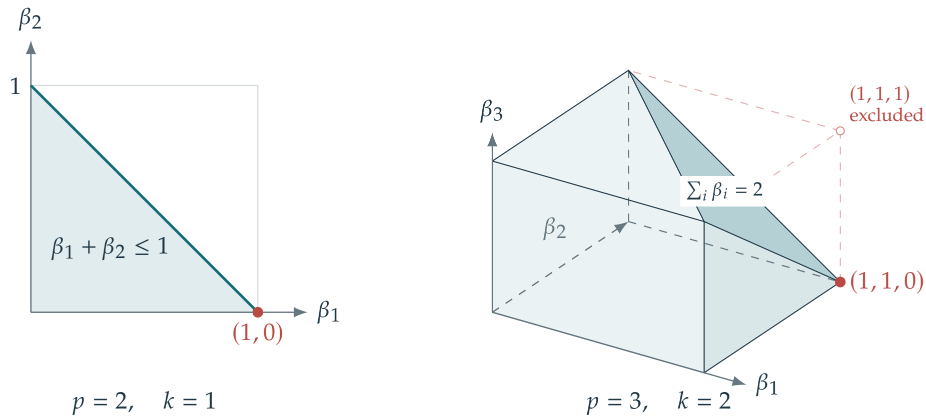
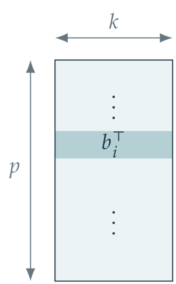
Panel 1



Feasible region: $0 \leq \beta_i \leq 1$, $\sum_i \beta_i \leq k$. Red dots: a maximizing vertex.

$$\lambda_1 \geq \cdots \geq \lambda_p \geq 0, \qquad \max \sum_i \lambda_i \beta_i = \sum_{i=1}^k \lambda_i$$

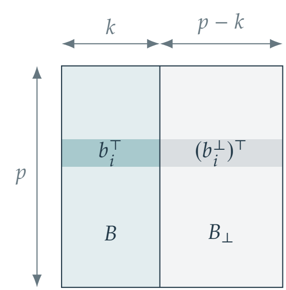
Panel 2



$$B = V^\top S, \quad B^\top B = I_k$$

$$\beta_i = \|b_i\|^2 \geq 0, \quad \sum_i \beta_i = k$$

Panel 3



$$\widetilde{B} = [B \; B_\perp], \quad \widetilde{B}\widetilde{B}^\top = I_p$$

$$\|b_i\|^2 + \|b_i^\perp\|^2 = 1 \quad \Rightarrow \quad \beta_i \leq 1$$